

Transcriptomics and Proteomics of the Minimal Synthetic Bacteria

Enguang Fu ^{*1,10}, Troy A. Brier ^{†1}, Jay E. Cournoyer¹, Benjamin R. Gilbert¹, Sage A. Glass², Yang-le Gao¹, Yanbao Yu³, Zane R. Thornburg^{1,4,5}, Karrie Goglin², Gavin John², Tiana Mamaghani², Sujata Shivakumar², Abner T. Apsley^{5,10}, Christopher J. Fields⁷, John I. Glass^{2,10}, Angad P. Mehta^{1,8,10}, and Zaida Luthey-Schulten ^{‡1,4,5,8,9,10}

¹Department of Chemistry, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA

²J. Craig Venter Institute, La Jolla, CA 92037, USA

³Department of Chemistry and Biochemistry, University of Delaware, Newark, DE 19716, USA

⁴Cancer Center at Illinois, University of Illinois at Urbana-Champaign, Urbana, IL 61801, USA

⁵Beckman Institute for Advanced Science and Technology, University of Illinois at Urbana-Champaign, Urbana, IL 61801, USA

⁷Roy J. Carver Biotechnology Center, Urbana, IL 61801, USA

⁸Carl R. Woese Institute for Genomic Biology, Urbana, IL 61801, USA

⁹Department of Physics, University of Illinois Urbana-Champaign, IL 61801, USA

¹⁰NSF Science and Technology Center for Quantitative Cell Biology, Beckman Institute, Urbana, IL 61801, USA

*These authors contributed equally to this work.

†These authors contributed equally to this work.

‡Corresponding author: zan@illinois.edu

1 Abstract

150 words

2 Introduction

Background

Very beginning: Jacob and Monod paper in 1961 on bacterial gene co-expression, regulation, stoichiometric protein synthesis

Minimal and synthetic genomes (OLD manuscript but reduced)

Progress in RNA sequencing techniques enables direct observation of gene co-expression (OLD manuscript but reduced); bio-informatics prediction

What we did and achieved

Operonal structure revealed by long-read RNA seq with quantification by short-read

Correlation with proteomics

Genetic motifs of transcription initiation, termination, Shine-Dalgarno and downstream RNA processing

Synthetic biology and genome minimization of Syn1 and Syn3A in the context of operonal transcription; integrated with proteomics

Through the integration of transcriptomics and proteomics data, we are able to identify XX number of novel peptides ??

3 Results

3.1 Global Co-transcriptions in Operons

How transcription and further RNA processing can complexity the transcriptome even of the reduced organism

FLNC reads clustered to isoforms: 2.6M by 10 bps to 205k clustered; Use 10 for visual of 50 for operon backbone; distribution of sharpness 5' and 3' ends

Find the longest RNA isoform for representative of operons; then merge and rescue to get total 480 operons

Statistics on operon size: operon can cover anti-sense genes; longest is rPtn operon; operon length distribution;

Promoter and terminator signatures: -10 box significant; -35 box not; terminator predicted as hairpins + polyU

Protein complexes in operons

tRNAs in operons

Put one operon here: DCW operon?

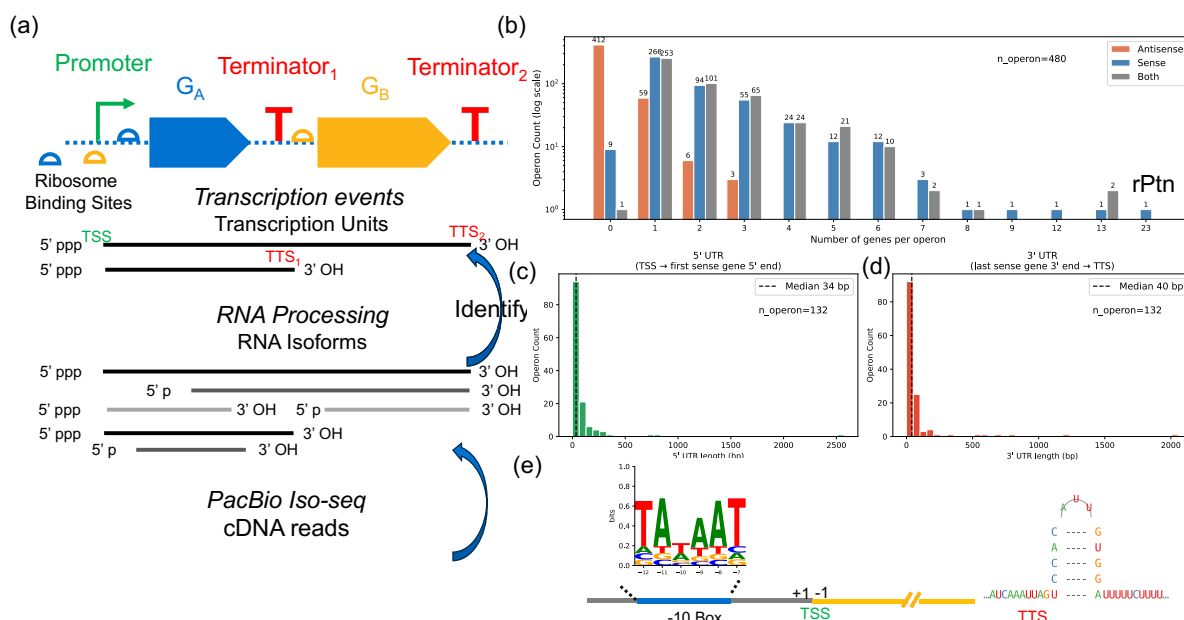


Figure 1: Caption

Extended Figure:

operon length distribution

-35 region of TSS: not characteristic

tRNA in operons

SI: Excel File Co-expression

Sheet 1 operon candidate;

Sheet 2 protein complexes: operon, expression level; protein abundance

Sheet 3 tRNAs

3.2 Prevasive and Biased RNA Processing Further Complexified the Transcriptome

Decay intermediates captured by full-length PacBio RNA sequencing: 5' erosion 0154/lap and 3' erosion 0178.

RNA isoform truncation categories: 25% of the isoforms are complete; biased 3' intragenic.

Could be Exo 3' to 5' initiate RNA degradation, unstructured 3' end to facilitate exo 3' to 5' digestion ; without experiments verification

Proposed mechanism to explain biased RNA processing: lower copy numbers of tmRNA/SmpB prevent the activities of RNase R on trapped mRNAs;

Intragenic truncated RNA isoforms strongly suggests endo-cleavage: ATP synthase operon have RNase III cleavage at *atpA* peripheral gene

Correlation protein localization and cleavage activities

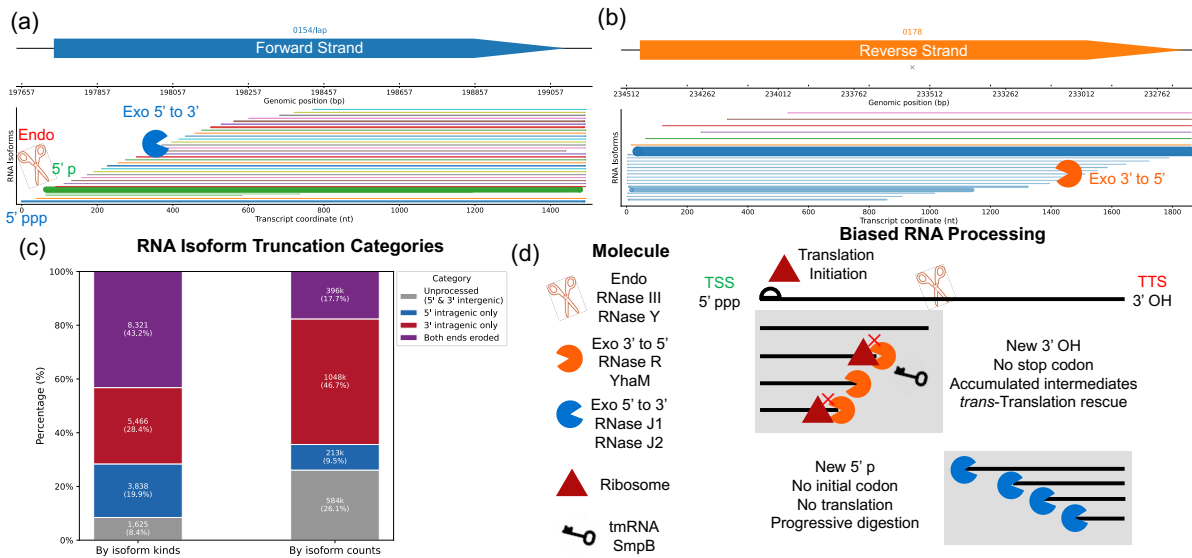


Figure 2: Caption

Table S1: Ribonucleases and *trans*-Translation Rescue System in Syn1

LocusNum	Gene Product	Abundance	Function	Activity Signature
<i>rnmV</i> /0003	RNase M5	37	Endo	5S rRNA
<i>rny</i> /0359	RNase Y	70	Endo	ssRNA
<i>rnc</i> /0418	RNase III	67	Endo	degenerate features
<i>rnjB</i> /0257	RNase J2	142	Exo 5'→3'	dsRNA
<i>rnjA</i> /0600	RNase J1	92	Exo 5'→3'	-
<i>yhaM</i> /0437	YhaM	117	Exo 3'→5'	5' monophosphate
<i>rnr</i> /0775	RNase R	36	Exo 3'→5'	3' OH
<i>ssrA</i> /0158	tmRNA	2.5% of non-rRNA	Ribosome rescue	unstructured region
<i>smpB</i> /0776	Small protein B	14	tmRNA cofactor	3' OH
				unstructured region

SI: Excel file Gene-level RNA Processing and Expression

Sheet 1: RNA Processing mapped from B. subtilis

3.3 Correlation between Transcriptomics and Proteomics in the Reduced Organism

How to calculate the absolute abundance? Protein density; **Distribution of absolute copy numbers of proteins:** most abundant one; median; how protein synthesis resource biased toward metabolism?

Low coverage of membrane proteins: dist of ptns in different locations; poor digestion by trypsin alone

Use illumina to do the quantification

Correlation coefficient of 0.67 found for cytosolic proteins: compare to other organisms; outliers; why those?

Then three levels to investigate the global deviation of proteomics from transcriptomics:

Translation initiation predicted by OSTIR failed to explain the deviation;

Elongation speed quantified by CAI dramatically increased the correlation factor: codon usage (AT rich genome);

Protein degradation speed mapped from Mpn significantly slower than the cell cycle of Syn1, thus not decisive on the protein abundance.

Extended Figure:

Dist of ptn copy numbers in different localization;

Illumina corr. w. PacBio

TIR w. residual

tAI w. residual

SI: Excel file Gene-level RNA Processing and Expression

Sheet 2: Transcriptomics and Proteomics of all syn1 genes

3.4 Novel Transcription and Translation Activities

2% of isoforms has anti-sense coverage with classified transcription schemes: length and read counts distribution mostly are minor expressed

Intergenic coverage higher due to the 5' and 3' UTRs and gap between genes for polycistronic operons.
Purely intergenic region

Synthetically inserted genes: water marks

how inserted genes disturb the expression of original genes?

Novel short peptide/protein revealed by mass spectrometry

RNAP bidirectional collision

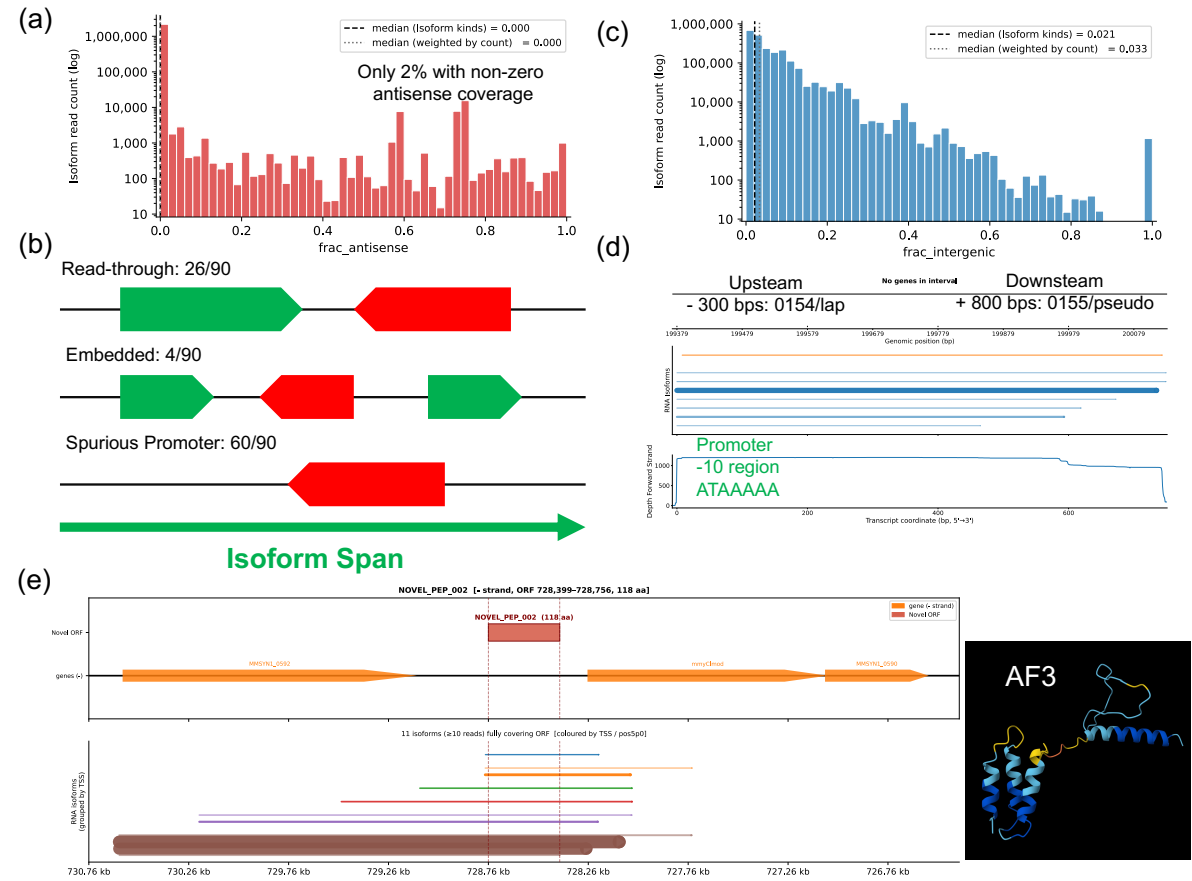


Figure 3: Caption

Extended Figure:
his/0918 mainly transcribed by anti-sense
Extra N termin of gene 0768/mmyCIVR

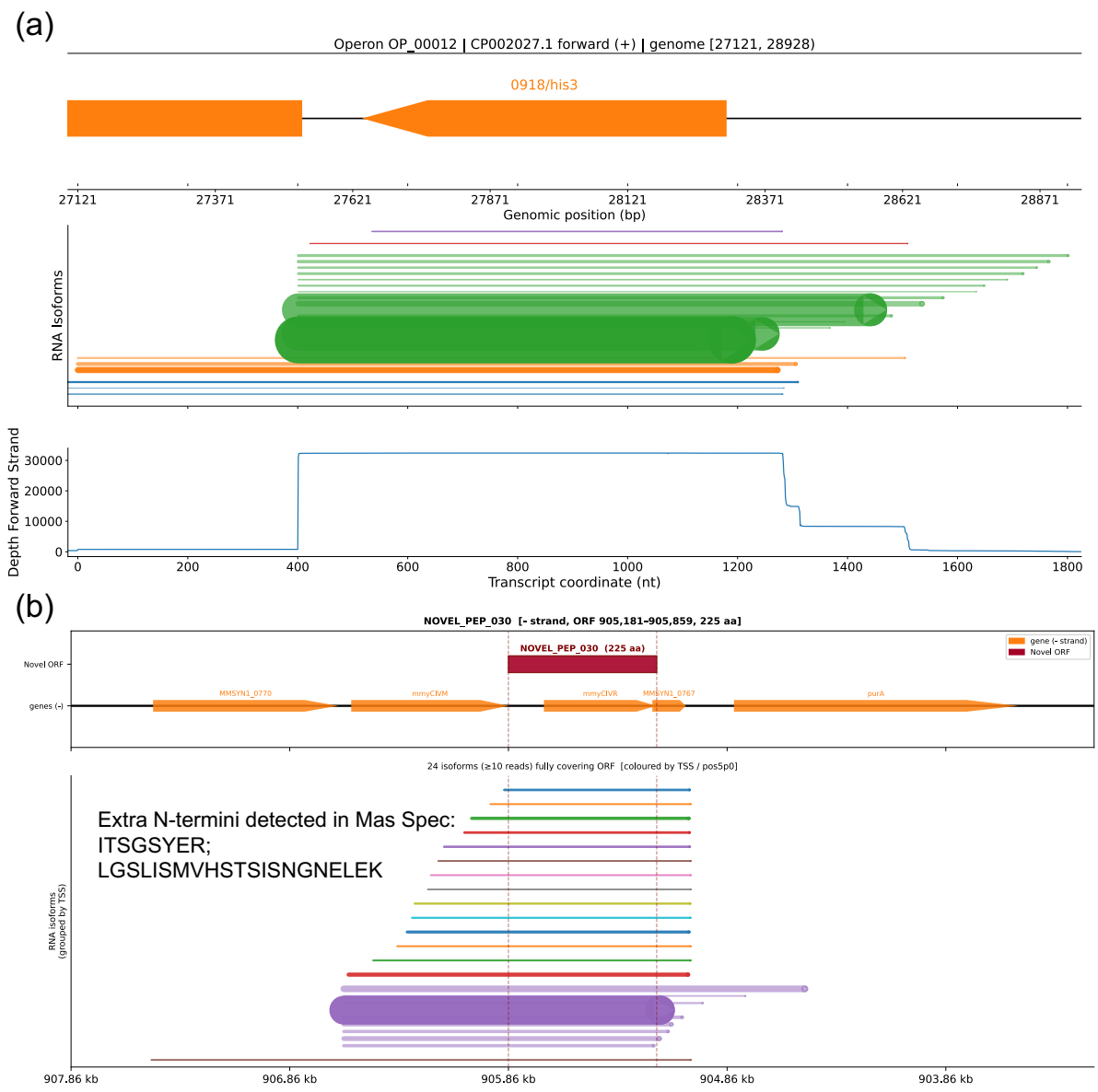


Figure 4: Caption

SI: Excel file Gene-level RNA Processing and Expression
Sheet 3: novel transcription and translation

3.5 Operonal Structure Changes to the minimal cell, JCVI-syn3A

3.6 Transcriptome and Proteome Changes Affected the Cell Living

revisit gene essentiality from gene co-expression

Statistics on how the operon broken

The genome reduction from syn1 to syn3A was systematically assessed one more time and visualized as **a pie chart**: # of deletion blocks, distribution of block lengths, insertion events.

We present the genome reduction's impact on transcriptome on two levels: mechanistically operonal structure, and the resultant gene expression TPMs.

Operon levels: intact operons;

operon pairs at deletion junctions: **newly fused gene coexpression**.

gene transcription context change upon deletion; **promoter lost genes has significant less expression compared to others genes**.

HupA as special case

Gene-level change: **mRNA pool composition: 400 mRNA coding genes deleted, occupy 20% of mRNA pool; unclear proteins only occupy 3%**;

TPM change plot: ribosomal proteins in two operons upgraded; central metabolism transcriptoin downgraded

Absolute TPMs: **ribosomal protein coding genes upgraded**

FC TPMs: Minorly expressed in syn1 but decently expressed in syn3A: majorities are still uncharacterized.

FC at RNA level: L31, hupA, thiB has $FC \leq 0.1$ in both ONT and Illumina; yet, L31 and thiB with increased protein level

genes upgraded at RNA level (26 genes with $FC_{RNA} \geq 10$): majority are uncharacterized protein that minimally expressed in Syn1 but slightly in Syn3A; iPM FC with medium of 2; non-linearity

0447/dut: increased to decent expressed; what happened to transcription; but iPM ≈ 0.17 ; PacBio TPM of 0447 much larger than Illumina; 0447 co-expressed with 0445, 0446 and 0448.

Translation efficiency

FC at protein level: 10 $FC \geq 10$; 8 out of 10 with iPM ≤ 100 ; rpmF with decreased RNA expression; 0115 and 0530 both upgraded at RNA and protein level

HupA decreased at both RNA and protein level;

XseAB decreased at both RNA and protein level

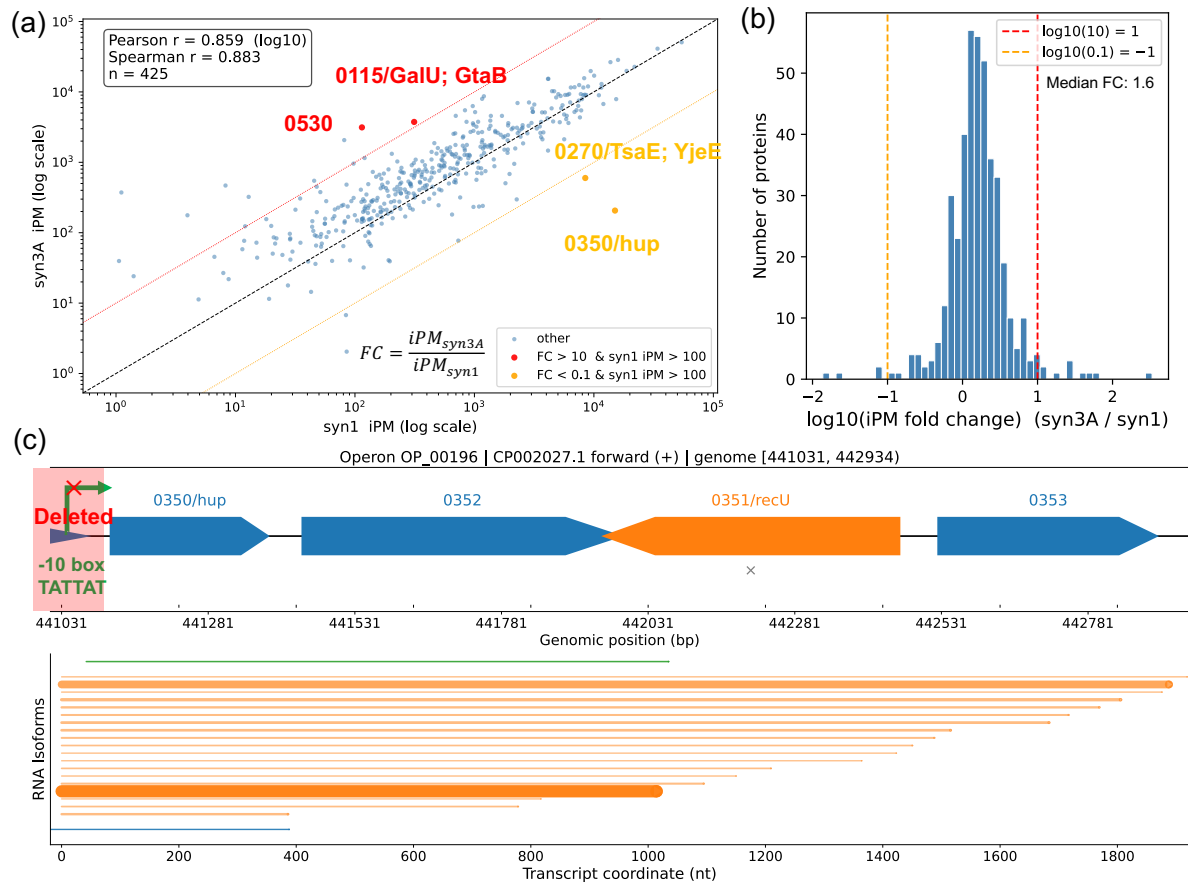


Figure 5: Caption

SI: Excel File Genome Reduction

Sheet 1: revisited gene essentiality

Sheet 2: syn1 vs syn3A

Sheet 3: syn3A proteomics and transcriptomics

4 Discussion

First genome-scale full-length transcriptomics of synthetic bacteria Syn1 and Syn3A

Complete and standard pipeline to infer operon from full-length RNA sequencing. Chris suggests publishing a toolkit. Why not since not so difficult to do and it will sell the paper better?

Leverage bioinformatics to largely explain transcription initiation, termination and the downstream RNA processing processes

Pave the road to simulate gene co-expression via RNA convoy on the bacterial circular genome

Hints on differential protein synthesis via polysomes translation of polycistronic transcripts

Ribosomal and membrane protein's abundances are less reliable than cytosolic ones -¿ Do ribosome profiling

5 Methods

5.1 Illumina Short-read Sequencing of Syn1.0 Transcriptome

5.1.1 RNA Sample Preparation

Sample preparation was conducted following [1] A frozen stock of Syn1.0 strain was provided by the JCVI. The frozen cells stock was scraped and inoculated in 10 mL of SP4+KnockOut media (media components listed in **Table ??**) under BSL2 conditions and incubated in a horizontal rotating shaker at 37°C for about 24 hours until the culture turned to media orange. The SP4 media is made from a two part formulation: part **I**—3.5 g Mycoplasma Broth Base, 10 g Bacto Tryptone, 5.3 g Bacto Peptone, and 600 mL distilled water mixed to suspend well, adjusted to pH 7.5, and then autoclaved 15 min at 121°C and part **II**—25 mL Glucose 20% w/v stock, 50 mL CMRL 1066 (10X stock w/o phenol red, w/o bicarb, w/o Gln), 14.6 mL sodium bicarbonate 7.5% w/v stock, 5 mL L-glutamine 200 mM stock, 35 mL Yeast extract solution, 100 mL TC Yeastolate 2% w/v stock, 170 mL Serum (heat inactivated FBS, HS) or substitute (KO), 2.5 mL Penicillin G (400,000 U/mL stock), and 1.5 mL Phenol red (1% w/v) filtered and sterilized (0.2 μ m) then stored at 4°C. To complete the SP4 medium, combine 1.5 volume of part **I** with 1 volume of part **II**.

RNA purification and depletion steps were conducted using sterile, RNase-free pipette tips (Thomas Scientific 1145N16, VWR 76322-134) and microcentrifuge tubes (Posi-Click 1149K01) on a countertop or biosafety cabinet treated with RNase decontamination solution (Invitrogen AM9780). Solutions apart from those provided with the kits were prepared using DEPC-treated water (Invitrogen 46-2224). RNA was extracted from JCVI-syn1.0 cells using the PureLink RNA Mini Kit (Invitrogen 12183018A) according to the manufacturer’s protocol. Briefly, $\leq 1 \times 10^9$ cells (2 mL of exponential phase culture) were harvested (4000 $\times$ g, 5 min, 4°C) and resuspended in lysozyme solution (100 μ L; 10 mM Tris-HCl, pH 8.0, 0.1 mM EDTA, and 20 mg/mL lysozyme Egg white (L-040-25, Goldbio). SDS solution (500 μ L; 10% w/v) was added and the cells were incubated for 5 min at room temperature (rt). Lysis buffer containing β -mercaptoethanol (350 μ L) was added and the sample was vortexed and passed six times through an 18-gauge needle. The resulting homogenate was centrifuged (12,000 $\times$ g, 2 min, rt) and 100% ethanol (250 μ L) was mixed with supernatant separated from the precipitate. The sample was purified through a spin cartridge provided with the kit and eluted with 2 $\times$ 100 μ L DEPC-treated water. To remove DNA, 1/9 volume of 10X DNase I Buffer (NEB, B0303S) and DNase I (NEB M0303S; 4 U total) were added and the mixture was incubated for 10 min at rt. EDTA (5 mM final concentration) was then added, and the reaction was heat-inactivated for 10 min at 75°C. Quantification with a NanoDrop Lite Spectrophotometer (Thermo Fisher ND-LITE-PR) typically showed yields of 40-60 μ g of RNA.

Ribosomal RNA was removed from the DNase-treated RNA samples using the RiboMinus Bacteria 2.0 Transcriptome Isolation Kit (Invitrogen A47335) according to the manufacturer’s protocol. Briefly, ≤ 5 μ g of DNase-treated RNA was mixed with 2X Hybridization Buffer (50 μ L) and RiboMinus Pan-Prokaryote Probe Mix (3 μ L) to a final volume of 100 μ L. The RNA/probe mix was denatured for 10 min at 70°C, followed by hybridization of the rRNA and probes for 20 min at 37°C. The RNA/probe mix was added to a microcentrifuge tube containing a 200 μ L suspension of RiboMinus Magnetic Beads in 1X Hybridization Buffer and incubated for 15 min at 37°C. The tube was placed on DynaMag-2 Magnet stand (Thermo Fisher 12321D) for 1 min. The supernatant (300 μ L) containing the rRNA-depleted RNA was carefully aspirated and mixed with Nucleic Acid Binding Beads (10 μ L), Binding Solution Concentrate (400 μ L), and 100% ethanol (1 mL). The mixture was incubated for 5 min at rt and placed on the magnetic stand for 5 min. The supernatant was discarded, the beads were washed with Wash Solution (300 μ L) and placed on the magnetic stand again. The supernatant was discarded, the beads were air-dried for 5 min, and the RNA was eluted with DEPC-treated water heated to 70°C. The Nucleic Acid Binding Beads were removed using the magnetic stand, and the supernatant was transferred to a fresh microcentrifuge tube.

5.1.2 RNA Sample Quality Control Analysis

Quantification of RNA was conducted using the Qubit RNA High Sensitivity assay (Invitrogen Q32852) and measured on a Qubit fluorometer (Invitrogen). Within-range samples treated with the RiboMinus kit were monitored with an AATI 5200 Fragment Analyzer (Agilent) showing depletion of rRNA-associated peaks. Method name: DNF-472T22 - HS Total RNA 15nt.mthds; Gel prime: No; Full conditioning: Yes; Gel prime to buffer: Yes; Gel selection: Gel 2; Perform prerun: 7.0 kV, 30 sec.; Rinse: No; Marker 1: No; Rinse: Tray: 3, Row: A, Dip count: 2; Sample injection: 6.0 kV, 150 sec.; Separation: 7.0 kV, 31.0

min.; Tray name: Tray-3; Analysis mode: RNA (Eukaryotic). Quality controls plots of RNA libraries have been included in ??, Fig. ??.

5.1.3 Illumina Library Preparation and Sequencing

The RNAseq libraries were prepared with the Kapa Hyper Stranded mRNA library kit (Roche). The library pool was quantitated by qPCR and sequenced on one MiSeq Nano flowcell for 251 cycles from each end of the fragments using a MiSeq 500-cycle sequencing kit version 2.

5.1.4 Illumina MiSeq Read Processing, Mapping to the Genome

Three strand-specific RNA-seq datasets of Syn1.0 were retrieved from the NCBI SRA database (accessions SRR35996296, SRR35996297, and SRR35996298), each comprising paired-end reads of 2×251 nt sequenced on an Illumina MiSeq. Datasets SRR35996296 and SRR35996297 are technical replicates of one biological RNA sample, and SRR35996298 is a second biological sample, yielding ~ 505 k, ~ 511 k, and ~ 1.09 M read pairs, respectively. Raw reads were assessed with **FastQC** and aggregated with **MultiQC** [2]; no adapter or quality trimming was applied prior to alignment. Reads were aligned to the Syn1.0 reference genome (NCBI accession CP002027.1) with **bowtie2** v2.5.5 [3] in default paired-end mode, against an index built with **bowtie2-build** from the same reference. Overall alignment rates were 99.49–99.56% across the three datasets, of which 89.4–93.1% of pairs aligned concordantly exactly once and 5.1–7.8% aligned concordantly more than once. SAM output was converted, sorted, and indexed with **samtools** v1.22.1 [4]. Strand-specific BAMs were generated using the dUTP / fr-firststrand flag convention (read 2 forward on the + strand and read 1 reverse on the + strand define plus-strand reads; their reverse complements define minus-strand reads), matching the Kapa Hyper Stranded mRNA library preparation, and per-strand sequencing-depth bedGraph tracks were computed with **samtools depth**.

5.1.5 Quantification of TPM from Sequencing Depth

Per-gene transcript abundances were quantified from the strand-specific depth tracks rather than from read counts. For each gene annotated in the Syn1.0 reference (**syn1.genes.gff3**, feature type **gene**), a sense and an antisense mean per-base depth were computed over the gene body from the plus- and minus-strand bedGraph tracks using prefix sums, with the sense strand taken as the gene’s annotated strand. Dividing the summed depth by the gene length, $\bar{d}_g = L_g^{-1} \sum_{i \in g} d_i$, length-normalizes coverage, and each value was scaled to transcripts per million as $\text{TPM}_g = 10^6 \bar{d}_g / \sum_{g'} (\bar{d}_{g'}^{\text{sense}} + \bar{d}_{g'}^{\text{antisense}})$, using a single sense-plus-antisense denominator per dataset. TPM was computed independently for the three datasets, which agreed well (Pearson $r = 0.98$ between the two technical replicates and $r = 0.92$ – 0.94 between biological samples). The datasets were merged by a two-step average, first across the technical replicates within their shared biological sample and then across the two biological samples at equal weight, yielding a representative Syn1.0 Illumina TPM profile (**syn1_illumina.TPM_profiles.tsv**).

5.2 PacBio Long-read Sequencing of Syn1.0 Transcriptome

5.2.1 RNA Sample Preparation

Sample preparation was conducted in the same manner as in **Section 5.1.1**, except DNase-I treated RNA was purified by spin cartridge (GeneJET RNA Purification Kit, Fisher K0731) instead of heat-inactivated with EDTA. In addition, prior to rRNA depletion, DNase I-treated RNA was polyadenylated using 1–10 μg of the RNA, ATP (1 mM final concentration) (NEB B0756), *E. coli* Poly(A) Polymerase (NEB M0276S), and *E. coli* Poly(A) Polymerase Reaction Buffer (B0276S). The reactions incubated at 37°C for 20 minutes and then purified using the GeneJET RNA spin cartridge.

5.2.2 PacBio Library Preparation and Sequencing

Ribodepleted and polyadenylated RNAs were purified using an RNA Clean & Concentrator-5 column (Zymo Research, Cat. #R1015) following the protocol designed to remove fragments smaller than 200 nucleotides. Twenty nanograms of purified RNA were converted into complementary DNA (cDNA) using the IsoSeq Express Oligo Kit from Pacific Biosciences. The cDNA molecules were barcoded with the Barcoded Overhang Adaptor Kit 8A and further processed into a library using the SMRTbell Prep Kit 3.0 (Pacific Biosciences). The resulting libraries were pooled at equimolar concentrations, and the pool

was sequenced on one SMRT Cell 8M using a PacBio Sequel IIe system in CCS sequencing mode with a 30-hour movie.

5.2.3 RNA Sample Quality Control Analysis

QC was conducted in the same manner as in **Section 5.1.2**. Quality controls plots of RNA libraries have been included in ??, **Fig. ??**.

5.2.4 PacBio Iso-Seq Read Processing, Mapping to the Genome, and Quality Control

Three technical replicates were generated for one biological sample of the synthetic bacterium JCVI-Syn1.0 (NCBI SRA accessions SRR36012641, SRR36012642, and SRR36012643). The three replicates were pooled into one read file of 2.95 M entries given their high consistency. Demultiplexed PacBio HiFi cDNA reads were preprocessed with a custom streaming pipeline before isoform clustering. The pipeline comprises four sequential steps: strand orientation, primer removal, 3' poly(A) trimming, and post-mapping quality control. Each step operates on FASTQ file; no third-party Iso-Seq tool (e.g. `lima`, `isoseq3 refine`) was used, giving explicit control over filter thresholds in this bacterial, non-natively-polyadenylated context.

Each double-stranded cDNA read carries one of two primer configurations: an antisense molecule reads 5'–H1 (oligo-dT handle, TAAGCAGTGGTATCAACGCAGAGTAC, 26 nt)–polyT–[reverse-complement of mRNA]–BCRC–3', whereas a sense molecule reads 5'–BC (barcode, CCCAACCCCTGCGACTTCATTGCA, 23 nt)–mRNA–polyA–H2–3'. A read was reverse-complemented to sense orientation if H1 occurred within its first 80 bp or if it ended in BCRC; otherwise it was kept as-is. Oriented reads were then required to terminate in an exact polyA (≥ 10 nt) + H2 suffix as evidence of a full-length sense cDNA. Of 2.95 M raw HiFi reads, 1.35 M were flipped via H1 and 191 k via BCRC; 255 k lacking the canonical polyA+H2 suffix were discarded, yielding 2.69 M oriented reads.

H2 was trimmed from the 3' end and BC from the 5' end with zero-mismatch tolerance (minimum overlap 18 bp); reads missing either primer were dropped. This removed a further 71 k reads, yielding 2.62 M primer-trimmed reads.

Syn1 transcripts are not natively polyadenylated, so any residual 3' A-run reflects the oligo-dT template-switch artifact and should be removed. Across the 2.62 M primer-trimmed reads the trailing-A run had median 30 nt and 99th percentile 34 nt, but a heavy tail reached 1.15 knt. Because the Syn1 genome contains no A-stretch longer than ~ 40 nt, runs > 70 nt cannot be biological and were discarded as likely concatemers or internal-priming artifacts; retained reads had up to 35 trailing As removed.

Cleaned reads were mapped to JCVI-Syn1.0 (CP002027.1) with `minimap2` v2.30 [5] and sorted and indexed with `samtools` v1.22.1 [4], producing 2.62 M primary alignments. Alignments were generated with the `-ax map-hifi` preset, which is optimized for high-accuracy PacBio HiFi reads and enforces contiguous read-to-genome alignment without splice junction inference.

A per-read QC table was then built from the sorted BAM with `pysam`, grouping each read's primary and supplementary alignments. Reads were failed if any of the following was violated: $\text{MAPQ} \geq 20$; primary aligned fraction ($\sum \text{of } M/I/=X \text{ over query length}$) ≥ 0.70 ; primary soft-clip fraction ≤ 0.30 ; $|\text{query length} - \text{primary reference span}| \leq 100$ bp. Reads were additionally flagged as strong concatemers when they produced ≥ 2 supplementary alignments that either covered the same locus with a $\geq 1.5\times$ reference-span ratio, or mapped to disjoint loci (different contig/strand, or > 200 bp genomic gap). All failing reads were removed from the final high quality (HQ) BAM.

99.3% of primary-mapped reads (2.62 M $\rightarrow$ 2.6 M) passed all QC thresholds and constitute the HQ BAM used for all downstream analyses. Read length is typical of bacterial mRNA isoforms (median 1.56 kb; 5th–95th percentile 0.88–3.07 kb; 99th percentile 3.93 kb; maximum 10.9 kb). Mapping quality is uniformly high (median MAPQ 60, mean 60.0, minimum 20 after filtering). The primary aligned fraction is essentially saturated (median 1.00, mean 0.998, 5th percentile 0.994), and the soft-clip fraction is negligible (median 0, mean 0.2%, 95th percentile 0.6%). 95% of retained reads have $|\text{query length} - \text{reference span}| \leq 10$ bp and 99% ≤ 55 bp, confirming near-gapless full-length alignments.

5.2.5 RNA Isoform Clustering

Isoform clustering of PacBio FLNC reads. 2.6 M primary FLNC alignments yielded 621 k unique (chrom, strand, pos5p0, pos3p0) tuples at single-bp resolution, with 77.3% supported by a single read — far exceeding the 911 annotated Syn1 genes and reflecting a combination of biological TSS/TTS micro-heterogeneity, alignment noise, and low-count degradation products.

A bare read-count threshold is unacceptably lossy: requiring $n \geq 10$ reads per single-bp tuple discards ≈ 874 k reads ($\approx 34\%$ of all primary reads), most of which sit on tuples directly adjacent to high-count tuples and represent biological wobble around a single underlying TSS/TTS rather than independent transcripts. Clustering is required to recover this signal for quantification and to deduplicate near-identical end positions for downstream analyses.

To choose a bp-tolerance (eps), cluster counts were swept over a range of tolerances on informative subsets (tuples with $n \geq 10$ and $n \geq 50$ reads). The cluster count decreased smoothly with increasing tolerance and showed an inflection between 10 and 15 bp before flattening beyond 20 bp. An eps of 10 bp was chosen, consistent with the typical end-position accuracy of PacBio CCS reads and the 10-bp deduplication tolerance used in the operon segmentation, and yielding biologically sensible counts (≈ 1 k distinct transcription units among ≈ 900 genes organized in operons).

Tuples were clustered per (chromosome, strand) by complete-linkage under Chebyshev distance with a cutoff of eps = 10 bp. Complete linkage (rather than single linkage) was required to prevent chain-like merging through the dense carpet of low-count degradation-product tuples in highly expressed operons, which would otherwise fuse multiple real TSSs into a single giant cluster. No minimum-read filter was applied at cluster construction, so the isoform catalog is complete and downstream analyses select their own thresholds. For each cluster, the read-count-weighted median 5'/3' end positions, the median absolute deviation (MAD) of member positions on each axis, and the total number of supporting reads were reported.

Complete-linkage clustering collapsed 621 k single-bp tuples into 267 k isoform clusters. The diameter guarantee was reflected in the empirical distributions: median MAD was 0 bp on both 5' and 3' ends, the 99th percentile was 5.0 bp (5') and 4.5 bp (3'), and the maximum was bounded at 5 bp (= eps/2). Applying minimum-read thresholds to clusters rather than raw tuples substantially increased the quantified signal: at $n \geq 10$, clustering retained ≈ 21 k isoforms covering ≈ 2.1 M reads (vs. ≈ 17 k tuples covering ≈ 1.7 M reads, +23%); at $n \geq 50$, clustering retained ≈ 4 k isoforms covering ≈ 1.8 M reads (vs. ≈ 3.3 k tuples covering ≈ 1.5 M reads, +22%). The $n \geq 50$ set was used for operon segmentation and the $n \geq 10$ set for transcript visualization in subsequent figures.

Each cluster was annotated against the Syn1 gene model (GFF3) as sense, antisense, intergenic, or mixed using 95%-of-length assignment rules with a 50-bp minimum gene-overlap threshold.

5.2.6 Quantification of TPM from Sequencing Depth

Per-gene sense and antisense TPM were computed from the PacBio HQ per-strand depth tracks (`syn1.PacBio.FLNC.HQ.{p}`) using the same depth-based definition applied to the Illumina data, namely the gene-length-normalized mean per-base depth scaled to transcripts per million against a single sense-plus-antisense denominator. Because the three technical replicates were pooled into a single library before mapping, no replicate averaging was applied, yielding one Syn1.0 PacBio TPM profile (`syn1_pacbio_TPM_profiles.tsv`).

5.3 Operon Identification from PacBio Transcriptomics of Syn1.0

5.3.1 Isoform-based operon segmentation

Of the 267 k clustered RNA isoforms, 4 k supported by ≥ 50 reads were retained to suppress noise from sequencing artifacts and RNase degradation fragments. These isoforms were grouped into candidate operons by containment clustering: two isoforms were placed in the same cluster if one was fully contained within the other (within a 10 bp boundary tolerance), implemented via a union-find algorithm with a coordinate sweep line. This produced 316 initial operons (148 on the plus strand, 168 on the minus strand; median length 2,100 bp), which were then annotated against the 911 canonical *Syn1* genes, of which 307 operons contained at least one sense gene (108 single-gene, 90 two-gene, and 109 with three or more genes).

Same-strand operon overlaps (30 pairs, 26 of which shared at least one gene) were resolved in two steps. First, each overlapping pair was evaluated by a transcription-direction rule: if the downstream operon's TSS fell in an intergenic region or within 10 nt upstream of the nearest gene start (i.e. a plausible 5' UTR), the pair was kept separate as genuine independent promoters (15 pairs); otherwise they were merged as RNase processing artifacts (15 pairs), reducing the operon count from 316 to 301. Second, two additional same-strand operon pairs sharing identical sense gene loci were merged via a transitive union-find procedure, yielding 299 isoform-derived operons.

After the initial segmentation, 200 of 911 genes (22.0%) remained uncovered by any isoform-derived operon. These were rescued through three complementary strategies: (i) 13 groups of consecutive un-

covered genes (gap ≤ 500 bp) were recovered by identifying spanning isoforms in the BAM file, (ii) the two rRNA operons (16S–23S–5S) were added manually, as their extreme secondary structure precludes faithful long-read capture, and (iii) the remaining 166 uncovered genes with non-zero spanning reads were added as single-gene transcription units. The final operon map comprised 480 operons (283 from isoform clustering, 14 merged, 2 gene-combined, 15 multi-gene rescue, and 166 single-gene rescue) and achieved complete sense-strand coverage of all 911 annotated *Syn1* genes (91.4% sense-only, 5.8% both sense and antisense, 2.7% antisense-only, 0% uncovered).

5.3.2 Locate TSS, TTS, and RNA Cleavage Sites from PacBio RNASeq

Questions here: can FLNC faithfully represent the ends of native transcripts?

the start and end locations of each read, which are then used to determine the covered genes for this each. The likely TUs are defined by the consecutive covered genes. For each TU, the TSSs and TTSs are defined by locating the possible motifs at the 5' and 3' of the covered genes that identified by bioinformatic predictions. The truncated RNA reads tell the possible cleavage sites by ribonucleases.

The endo RNA cleavage sites are differentiated from the internal promoters based on the predicted and inferred locations of internal promoters.

Iso-Seq HQ consensus isoforms constitute a non-redundant set of unique transcript species derived from read-level data; while they define transcript boundaries and connectivity, they do not preserve molecule counts or relative abundances.

5.4 RNA Processing Analysis

5.4.1 Ribonucleases in *Syn1* and *Syn3A*

5.4.2 Homology Comparison of RNase III and RNase Y between *B. subtilis* and *Syn1*

Table S2: RNase III and RNase Y Homology Comparison

B. subtilis ^a		Syn1		Alignment	
Length	Domain ^b	Length	Domain	Coverage %	Identity %
249	RNase_III and dsDNA binding	232	RNase_III and dsDNA binding	91	45
520	RNase_Y, KH, and HD	509	RNase_Y, KH, and HD	98	35

^a NCBI genome accession B_subtilis_168_NC_000964.3

^b Inferred by InterPro

5.4.3 Mapping Endo-RNase Processing Sites from *B. subtilis* to *Syn1*

5.5 Proteomics of *Syn1* and *Syn3A*

5.5.1 Sample Preparation – Cell lysis and Protein digestion

Cell pellets of *Syn1.0* and *Syn3A* strains were mixed with 100 μ L of SDS-based lysis buffer (4% SDS, 100 mM Tris-HCl, pH 8.0) containing final concentration of 10 mM Tris (2-carboxyethyl) phosphine (TCEP) and 40 mM chloroacetamide (CAA), vortexed for 5 min, and then boiled at 95°C for 10 min. The lysate samples were then processed for proteomics analysis following the E3technology procedure reported recently [6]. In brief, the lysates were mixed with 4 $\times$ volume of 80% acetonitrile (ACN) and then transferred to E3filters (CDS Analytical, Oxford, PA) followed by centrifugation at 3,000 rpm for 2 min. The filters were washed with 500 μ L of 80% ACN for three times, and then transferred to clean collection tubes. For protein digestion, around 1.0 μ g of trypsin and 200 μ L of digestion buffer (50 mM triethylammonium bicarbonate, TEAB) were added to each sample followed by incubation at 37°C for 16-18 hours with gentle shaking (300 rpm/min). After digestion, the filters were first spun at 3,000 rpm for 2 min to collect the flow through, and then eluted sequentially with 200 μ L of 0.2% formic acid (FA) in water and 200 μ L of 0.2% FA in 50% ACN. The pooled elution was dried in SpeedVac, and stored under -80°C until further analysis.

5.5.2 LC-MS/MS analysis and protein identification

The LC-MS/MS analysis was performed using an Ultimate 3000 RSLCnano system in conjunction with an Orbitrap Eclipse mass spectrometer and FAIMS Pro Interface (Thermo Scientific), similar to the procedures described recently [7]. In brief, the peptides resuspended in LC buffer A (0.1% FA in water) were first loaded onto a trap column (PepMap100 C18, 300 $\mu\text{m} \times 2\text{ mm}$, 5 μm particle, Thermo Scientific), and then separated on an analytical column (PepMap100 C18, 50 $\text{cm} \times 75\text{ }\mu\text{m}$ i.d., 3 μm ; Thermo Scientific) at a flow of 250 nL/min. A linear LC gradient was applied from 1% to 25% mobile phase B (0.1% formic acid in acetonitrile) over 125 min, followed by an increase to 32% mobile phase B over 10 min. The column was washed with 80% mobile phase B for 5 min, followed by equilibration with mobile phase A for 15 min. For MS analysis, the Detector type was Orbitrap with resolution of 120,000; Precursor MS range (m/z) was 380–980; AGC target was Standard; Maximum injection time mode was Auto. For data-independent acquisition (DIA) MS/MS analysis, the Isolation mode was Quadruple; DIA Window type was Auto and Isolation Window (m/z) was 8 with an overlap of 1; activation type was HCD with fixed collision energy mode (30%); the Detector Type was Orbitrap with a resolution of 30,000; Normalized AGC target (%) was 800, and the Maximum injection time mode was Auto; the Loop Control was 2 sec. For FAIMS compensation voltages (CV) setting, a 3-CV combination (~ 40 , ~ 55 , and ~ 75) was applied.

For proteome identification, the MS raw data were processed using Spectronaut software (version 19), a library-free DIA analysis workflow with directDIA+ and the in-house curated protein databases (Syn1.0, 828 sequences; Syn3A, 458 sequences). Detailed parameters for Pulsar and library generation include: Trypsin/P as specific enzyme with minimum peptide length of 7 amino acids and maximally 2 missed cleavages; toggle N-terminal M turned on; Oxidation on M and Acetyl at protein N-terminus as variable modifications; Carbamidomethyl on C as fixed modification; FDRs at PSM, peptide and protein level all set to 0.01; Quantity MS level set to MS2, and cross-run normalization turned on. The MS raw data associated with this study has been deposited onto the MassIVE Repository with the accession MSV000099558 (password: 4zzHbHHqW7G12F5).

5.5.3 Relative Protein Quantification

Protein abundances were quantified from the Spectronaut iBAQ (intensity-based absolute quantification) values, which serve as a within-sample proxy for molar protein amount. For each of the three replicates per organism, iBAQ was converted to a relative abundance in parts per million (iPM), $\text{iPM}_i = 10^6 \text{iBAQ}_i / \sum_j \text{iBAQ}_j$, and the per-protein mean iPM across replicates was used for all downstream analyses, with the across-replicate coefficient of variation retained as a quality metric. For Syn1.0, 721 of the 828 annotated protein-coding genes (87.1%) carried a quantified iPM.

5.5.4 Absolute Intracellular Protein Quantification

The Syn1.0 mass spectrometry data provides relative abundances for 721 of the 828 protein-coding genes, which can be converted to absolute abundances by scaling the relative amount with the total number of proteins in the cell. The total number of proteins has not been measured directly within Syn1.0, but can be estimated via an approach similar to that used in [8]. First the dry mass of the cell and protein mass fraction are needed. The protein mass fraction is assumed to be 58.2%, which was measured in the parent organism, *Mycoplasma mycoides* subsp. *capri* [9]. The dry mass of the cell, 12.8 fg, was also estimated identically to the dry mass of JCVI-syn3A (see [8], *Appendix 1*). Accordingly, the average JCVI-syn1.0 cell has approximately 127,000 proteins; the cell volume is calculated as the sphere of 220 nm radius as 0.044 μm^3 [10]. Thus, a protein density of $2.87 \times 10^6 \text{ proteins}/\mu\text{m}^3$ is estimated, which is in rough agreement with $3.5 - 4.4 \times 10^6 \text{ proteins}/\mu\text{m}^3$ measured in *E. coli* [11]. Finally, absolute protein abundances were calculated using the following formula:

$$\text{Abs Abundance}_i = \frac{\text{dry mass} * \text{mass fraction}}{\text{avg molecular weight}} * \text{Rel Abundance}_i \quad (1)$$

5.5.5 Localization of Proteome

Subcellular localization was predicted from sequence for every Syn1.0 protein, since membrane-associated proteins are systematically under-recovered by the trypsin-digestion workflow and are therefore handled separately in the transcriptome–proteome correlation. Transmembrane regions (TMRs) were predicted with DeepTMHMM [12] and signal peptides with SignalP 6.0 [13]. Each protein was assigned to one of four

classes by priority: a SignalP secretory or lipoprotein signal peptide marked it as extracellular or lipoprotein, respectively; otherwise one or more DeepTMHMM TMRs marked it as membrane, and proteins with no TMR were assigned cytoplasmic. Of the 721 detected Syn1.0 proteins, 516 were cytoplasmic, 126 membrane, 68 lipoprotein, and 11 extracellular. Median copy number per cell fell steeply with membrane association, from 47 for cytoplasmic proteins to 21 (lipoprotein), 10 (membrane), and 3 (extracellular), consistent with the under-digestion and under-recovery of membrane-embedded proteins rather than a true abundance difference.

5.6 Correlation Between Transcriptome and Proteome

Gene-level proteome abundance (relative iPM) was correlated against transcriptome abundance (sense TPM) to test how faithfully transcript level predicts protein level in Syn1.0, and the deviations from this relationship were then attributed to translation and protein-turnover effects. The two layers were joined by locus tag (`Transcription.Translation.py`, output `syn1_genes.transcriptomics.proteomics.csv`), and the correlation was restricted to the 717 mRNA genes with non-zero signal in both. Illumina sense TPM was used as the transcriptome axis because it correlated more strongly with iPM than PacBio TPM did. On a $\log_{10}$ scale, transcript and protein levels were positively correlated across all detected proteins (Pearson $r = 0.61$, $r^2 = 0.38$, $n = 717$) and more tightly for the cytoplasmic subset (Pearson $r = 0.70$, $r^2 = 0.49$, $n = 512$), consistent with the under-recovery of membrane, lipoprotein, and extracellular proteins by the digestion workflow. For each gene the residual was defined as the offset of $\log_{10}(\text{iPM})$ from the ordinary-least-squares fit on $\log_{10}(\text{TPM})$, so that a positive residual marks more protein than the transcript predicts (efficient translation or a long-lived protein) and a negative residual marks less. Three layers, translation initiation, translation elongation, and protein degradation, were evaluated for their ability to explain this residual.

5.6.1 Translation Initiation Rate Prediction

Per-gene translation initiation efficiency was predicted from sequence to test whether initiation rate accounts for the transcriptome–proteome residual. For each detected mRNA gene, the PacBio isoforms (read count ≥ 10) whose transcribed span covered the full ORF were identified, and an initiation window spanning 30 nt of 5′ UTR (or the available UTR, whichever was shorter) through 30 nt into the coding sequence was extracted from the Syn1.0 genome with circular wraparound. Each window was scored with OSTIR [14] using the Syn1.0 16S rRNA anti-Shine–Dalgarno tail (ACCUCUUU), taking the prediction at the annotated start codon, and the translation initiation rate (TIR) together with its thermodynamic sub-terms (Shine–Dalgarno, standby-site, spacing, and 5′ mRNA-folding ΔG) was recorded per isoform. A read-count-weighted mean TIR was then computed for each gene, yielding a valid TIR for 566 genes. The contribution of initiation was quantified by adding $\log_{10}(\text{TIR})$ to the baseline ordinary-least-squares model $\log_{10}(\text{iPM}) \sim \log_{10}(\text{TPM})$ (baseline $r = 0.59$, $R^2 = 0.35$ on this subset) and by correlating $\log_{10}(\text{TIR})$ against the baseline residual. Translation initiation explained little of the residual: it raised the model R^2 by only 0.020 (2.0% of the variance), and $\log_{10}(\text{TIR})$ correlated weakly with the residual (Pearson $r = 0.17$, $P = 4 \times 10^{-5}$).

5.6.2 Codon Adaptation Index (CAI)

Codon optimality was quantified as a proxy for translation elongation efficiency and tested against the residual. A codon adaptation index (CAI) [15] was computed for all 828 frame-valid Syn1.0 coding sequences under the Mycoplasma genetic code (UGA = Trp), with the reference set of highly expressed genes taken as the top 20% of protein-coding genes by mean iPM (144 of the 721 detected genes, 91% of them cytoplasmic). CAI was essentially independent of transcript level (CAI vs $\log_{10}(\text{TPM})$, Pearson $r = 0.006$, $P = 0.87$) yet correlated with the proteome residual (Pearson $r = 0.36$, $P = 4 \times 10^{-23}$), carrying information nearly orthogonal to the transcriptome term. Adding CAI to the baseline model raised the explained variance from $R^2 = 0.38$ to 0.46 ($\Delta R^2 = +0.080$, a 21% gain over the baseline; $n = 717$), and the improvement held within the cytoplasmic subset (R^2 from 0.49 to 0.54, $\Delta R^2 = +0.052$; $n = 512$).

5.6.3 Protein Degradation Rate Mapping from Mpn

Syn1.0 protein half-lives have not been measured, so intrinsic degradation rates were transferred from the related minimal bacterium *Mycoplasma pneumoniae*, for which proteome-wide half-lives are avail-

able [16]. Syn1.0 and *M. pneumoniae* proteins were matched by reciprocal-best-hit orthology (reciprocal **blastp**, retaining pairs with $\geq 25\%$ identity, $\geq 50\%$ coverage on both proteins, and bitscore ≥ 50), and the *M. pneumoniae* half-life was assigned to each of the 245 Syn1.0 proteins (225 cytoplasmic, 19 membrane, 1 lipoprotein) that had a confident ortholog. Because turnover is set by the responsible protease, each transferred half-life was scaled by the *M. pneumoniae*-to-Syn1.0 abundance ratio of that protease, applying Lon to cytoplasmic proteins and FtsH to membrane and lipoproteins, with protease copy numbers and cell volumes taken from this study and from *M. pneumoniae* [17] (scaling factors 0.84 for Lon and 2.08 for FtsH). The shortest Syn1.0-corrected half-life was 4.7 h and the median was 32 h, all far longer than the ~ 1 h cell-doubling time, so protein loss is dominated by dilution through growth rather than by active degradation. Accordingly, adding $\log_{10}(t_{1/2})$ to the baseline $\log_{10}(\text{iPM}) \sim \log_{10}(\text{TPM})$ model increased R^2 by at most 0.01, with a half-life coefficient far below the value of 1 expected if abundance scaled with stability, and half-life was uncorrelated with the residual (Pearson $r = -0.12$, $P = 0.06$, $n = 242$).

5.7 Novel Transcription and Translation

5.7.1 Anti-sense and Intergenic Transcription

Each PacBio isoform cluster was labelled base-by-base against the canonical Syn1.0 gene model, and its sense, antisense, and intergenic coverage fractions were computed. Antisense coverage was carried by 2% of the isoforms (345 of those with ≥ 10 supporting reads), which collapsed to 90 representative clusters after merging near-identical ends within 10 bp. These were classified by the arrangement of their sense and antisense segments into three cases: antisense-initiated transcripts from a spurious promoter in the AT-rich genome (anti $\rightarrow \dots$), sense transcripts reading through into an antisense gene at an operon boundary (sense $\rightarrow$ anti), and read-through transcripts with an antisense gene embedded between sense genes (sense $\rightarrow$ anti $\rightarrow$ sense). Spurious-promoter transcripts accounted for 60 of the 90 clusters (67%), and read-through transcripts, including 4 embedded cases, for the remaining 30 (33%).

5.7.2 Novel Open-Reading Frames from Full-length RNA Isoforms

The abnormal isoforms, defined as those with >10 supporting reads and an abnormal fraction ($f_{\text{antisense}} + f_{\text{intergenic}} > 0.2$), gave 816 isoforms as input to the ORF-prediction pipeline.

All candidate start codons within the 816 isoforms were evaluated using OSTIR [14], which predicts translation initiation rates (TIR) by modelling Shine–Dalgarno (SD) complementarity to the Syn1.0 16S rRNA anti-SD (ACCUGCUUU) and local mRNA secondary structure near each AUG. For every predicted start site, the downstream reading frame was extended to the first in-frame stop codon (TAA or TAG) under genetic code 4, producing $\sim 26,000$ candidate ORFs (≥ 15 amino acids).

ORFs overlapping known sense reading frames were excluded, and the remainder were ranked by a synthesis rate score (isoform read count $\times$ TIR). The top 100 ORFs (of which $\sim 40\%$ exhibited strong SD-mediated initiation) were deduplicated by amino acid sequence and subjected to *in silico* trypsin digestion (one missed cleavage; peptides of 7–25 residues). Comparison of the resulting tryptic peptides against the canonical Syn1.0 proteome identified 44 novel ORFs producing at least one proteotypic peptide suitable for detection by bottom-up mass spectrometry. These 44 novel ORFs were appended to the Syn1.0 protein database, and the proteomics raw data were re-searched with Spectronaut.

5.8 Oxford Nanopore Long-read Sequencing of Syn3A Transcriptomes

5.8.1 ONT RNA Sample Preparation

One mL aliquots of actively growing JCVI-syn3A cells were placed in 25 mL of SP4 + 17% KnockOut serum replacement in 50 mL conical tubes and incubated overnight at 37°C. The cells were harvested in mid-late log phase as indicated by a red-orange color of the phenol red pH indicator in the growth media. The cultures were harvested when the media pH was 7.0. JCVI-syn3A cultures achieved this growth stage in 22 hours. Growth was arrested in the bulk cultures by placing the tubes on wet ice for 10 minutes. Keeping all tubes on ice, 1.4 mL of bulk cultures were distributed into fifteen 1.8 mL microfuge tubes per culture. These were centrifuged at 4°C for five minutes at 10,000 $\times$ g using a tabletop refrigerated centrifuge to pellet the cells. The supernatants from ten of the fifteen tubes were aspirated. In the remaining five tubes, the supernatant was then used to suspend pellets. The cell suspensions from those five tubes were then used to resuspend the pellets in five other tubes, and those more concentrated cell suspensions were used to resuspend the cells in the last five tubes. The JCVI-syn3A cells were now

in five tubes containing concentrated cell suspensions. These tubes were centrifuged for five minutes at 4°C under 10,000×g with a tabletop refrigerated centrifuge. All supernatants were aspirated completely, and the remaining pellets were snap frozen using liquid nitrogen. Sample tubes were transferred to dry ice and immediately stored at -80°C.

In the next step, cellular RNA was extracted from JCVI-syn3A cell samples using a phenol-chloroform extraction protocol based off procedures adapted from [1] and the ThermoFisher TRIzol™ Reagent user guide [18]. Briefly, 1 mL of 4°C TRIzol™ was added to each tube containing frozen cell pellets. The liquid was pipetted up and down multiple times to break up the cell pellet. The tubes were incubated at room temperature for 5 minutes to allow complete disassociation of nucleoprotein complexes. 200 µL of chloroform was then added and the tubes were mixed by shaking for 30 seconds. After the tubes incubated at room temperature for 3 minutes, they were centrifuged at 12,000×g for 15 minutes at 4°C. The upper RNA-containing aqueous phase was pipetted without collecting any of the white interphase separating the aqueous from the phenol-chloroform lower phase, and placed in fresh 1.8 mL tubes already containing 500 µL isopropanol. The tubes were incubated on ice for 10 minutes and then centrifuged at 12,000×g for 10 minutes at 4°C. The supernatant was aspirated and discarded off the white gel-like pellets. The tubes were inverted and the pellets were allowed to air dry for 15 minutes. The RNA pellets were then dissolved in 20 µL of RNase-free water containing 0.1 mM EDTA. RNA concentration was assessed with a Qubit RNA HS fluorometer assay and an Agilent TapeStation 2200. The next step was ribosomal RNA (rRNA) depletion.

The JCVI-syn3A samples were rRNA depleted using the NEBNext® rRNA Depletion Kit for bacteria. One µg of RNA was brought up to 11 µL with RNase-free water. To this were added 2 µL of NEBNext® Bacterial rRNA Depletion Solution and 2 µL of NEBNext® Probe Hybridization Buffer. The reaction was heated to 95°C for 2 minutes and cooled to 22°C at a ramp rate of 0.1°C/sec. This was followed immediately by RNase H digestion, where 2 µL of RNase H Reaction Buffer and 2 µL of NEBNext Thermostable RNase H were added to the hybridized RNA solution to a total volume of 20 µL. This reaction was incubated at 50°C for 30 minutes.

Following a bead cleanup with Agencourt RNAClean XP beads (Beckman Coulter™, cat # A63987), the 15 µL of the rRNA solution was Poly(A) tailed at the 3' end. RNA samples were mixed with a premade solution containing 2 µL of 10X *E. coli* Poly(A) Polymerase Reaction Buffer, 2 µL of ATP and 1 µL of the *E. coli* Poly(A) Polymerase (New England Biolabs) and incubated at 37°C for 30 minutes. The reaction was stopped by adding EDTA to a final concentration of 10 mM. Following another bead cleanup along with necessary quality control analysis on the samples, RNA was normalized to 9 µL containing 50 ng for JCVI-syn3A RNA samples.

5.8.2 RNA Sample Quality Control Analysis

For the JCVI-syn3A samples, both a Qubit RNA HS fluorometer assay and an Agilent TapeStation 2200 were used for assessing RNA concentration post-extraction from minimal cells. Initial Qubit values post RNA extraction were 1890 ng/µL for JCVI-syn3A. The Agilent TapeStation 2200 was used throughout the extraction and rRNA depletion steps to assess the quality of the RNA and depletion. TapeStation results gave an estimate of depletion through the absence of 16S and 23S peaks (Figure SXX). The corresponding graphs can be found in rRNA Supplemental Data. A Qubit RNA HS fluorometer was used to detect final quantitation of RNA.

Post poly-adenylation of RNA transcripts, the extent of the DNA and buffer contamination in each of the RNA samples was tested by performing standard spectroscopic measurements (NanoDrop One) and using the Qubit 1× dsDNA HS assay kit (ThermoFisher Scientific). Input RNA samples for Oxford Nanopore Direct RNA Sequencing library preparation were finally quantified using the Qubit RNA HS assay kit. RNA concentration values were used to normalize samples to the desired input volume of 9 µL containing 50–500 ng of input RNA for library preparation.

5.8.3 ONT Library Preparation and Direct Sequencing

After normalization, libraries for Oxford Nanopore sequencing were prepared using the ONT Direct RNA Sequencing Kit (SQK-RNA002) with slight modifications. Specifically, 0.5 µL of the RT adapter enzyme (reduced from 1 µL) and 4 µL of the RMX RNA adapter (reduced from 6 µL) were used during library preparation for JCVI-syn3A. The samples were sequenced using R9.4 flow cells on the ONT GridION platform. Live base-calling was enabled through the recommended MinKNOW (v23.07.5) scripts to generate fast5 files. 734.08 k reads were generated for JCVI-syn3A over a single run, with an average N50 of 430.

5.8.4 Processing of Raw ONT Reads

Results of the Nanopore reads were collected as individual fast5 files. The individual Nanopore reads were then combined into multi-read files for easier data analysis. Current signals were converted to canonical bases by base-calling. For the JCVI-syn3A experiments, base-calling was performed with Guppy (v7.0.9) using the RNA trim strategy and with calibration detection and filter enabled. In all run reports, base-called bases were 99.99–100% confident.

5.8.5 Mapping ONT Reads to Syn3A Genome

Reads were aligned to the JCVI-syn3A reference genome (NCBI accession CP016816.2) using `minimap2` v2.30 [5] with the long-read Oxford Nanopore preset for non-spliced alignment: `-ax map-ont -p 0.99 --MD`. The `map-ont` preset was selected in preference to the splice-aware preset commonly used for ONT direct RNA because JCVI-syn3A, like its parent *Mycoplasma mycoides* subsp. *capri*, is intronless; allowing N-skip (spliced) CIGAR operations would yield single-read alignments that bridge tens to hundreds of kilobases of unrelated reference. Strandedness of direct-RNA reads is preserved through the BAM flag bits. Resulting SAM output was converted, sorted, and indexed using `samtools` v1.22.1 [4].

Of 734 k input reads, 559 k (76.2%) mapped as primary alignments to the syn3A genome, with a mean read length of 383 nt (range 49–2,858 nt) and mean per-read Phred quality of 31.2; 98.2% of primary mapped reads exceeded a mean quality of Q20. The 175 k unmapped reads (23.9%) were of substantially lower quality (mean Q20.7), with 42.0% falling below Q15 and 61.0% shorter than 300 nt, consistent with the expected tail of low-quality and truncated molecules characteristic of ONT direct RNA sequencing rather than reference mismatch. Secondary alignments accounted for 3.6% of primary mapped reads (20.2 k alignments) and clustered tightly at two genomic loci (~55,460 and ~343,267 bp), corresponding to the two ribosomal RNA operons of syn3A. A further 7.3 k supplementary alignments (split-read mappings) were retained.

5.8.6 Mapping Illumina Reads to Syn3A Genome

Raw Illumina reads (paired-end, 51 nt, ~12.7 M pairs per replicate) of Syn3A wild type at log-phase from [19] were retrieved from NCBI SRA database and assessed with `FastQC` and aggregated with `MultiQC` [2]. Per-base and per-read quality scores passed all checks and no Illumina adapters were detected, so no trimming was applied.

Reads were aligned to the JCVI-syn3A reference (NCBI accession CP016816.2) with `bowtie2` v2.5.5 [3] in default paired-end mode. The overall alignment rate was 98.88%: 83.55% of pairs mapped concordantly exactly once, 10.43% concordantly with >1 hit (corresponding to the two ribosomal RNA operons of syn3A), and 6.02% of pairs were discordant or unpaired. SAM output was converted, sorted, and indexed with `samtools` v1.22.1 [4]. Strand-specific BAMs were generated using the dUTP / fr-firststrand flag convention (read 2 forward on + strand and read 1 reverse on + strand define plus-strand reads; their reverse complements define minus-strand reads), matching the Kapa Stranded RNA-Seq Kit library preparation, and per-strand sequencing-depth bedGraph tracks were computed with `samtools depth`.

5.8.7 Quantification of TPM from Sequencing Depth

Per-gene sense and antisense TPM were computed for Syn3A from the strand-specific depth tracks, using the same depth-based definition applied to the Syn1.0 Illumina data, namely a gene-length-normalized mean per-base depth scaled to transcripts per million against a single sense-plus-antisense denominator per dataset. Genes were taken from the Syn3A annotation (`syn3a_genome.gff3`), parsing both the `gene` and `pseudogene` feature types so that the three annotated pseudogenes were retained, for a total of 496 loci. TPM was computed independently from the Illumina and the ONT depth tracks; the single biological replicate of each platform required no replicate averaging, and the two profiles were merged into one table (`syn3A.TPM.Illumina.ONT.tsv`). The Illumina profile served as the quantitative transcript-abundance track, with the long-read ONT profile retained as an orthogonal check, and it reproduced the per-gene TPM reported for the same wild-type Syn3A library by [19] (Pearson $r = 0.998$ on $\log_{10}$ TPM), validating the depth-based pipeline.

5.9 Genome Reduction

5.9.1 Protein Fold of Change

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References

- [1] Grünberger, F, Reichelt, R, Bunk, B, Spröer, C, Overmann, J, Rachel, R, Grohmann, D, Hausner, W (2019) Next Generation DNA-Seq and Differential RNA-Seq Allow Re-annotation of the *Pyrococcus furiosus* DSM 3638 Genome and Provide Insights Into Archaeal Antisense Transcription. *Front. Microbiol.* 10.
- [2] Ewels, P, Magnusson, M, Lundin, S, Käller, M (2016) MultiQC: summarize analysis results for multiple tools and samples in a single report. *Bioinformatics* 32:3047–3048.
- [3] Langmead, B, Salzberg, SL (2012) Fast gapped-read alignment with Bowtie 2. *Nature Methods* 9:357–359.
- [4] Li, H, Handsaker, B, Wysoker, A, Fennell, T, Ruan, J, Homer, N, Marth, G, Abecasis, G, Durbin, R, 1000 Genome Project Data Processing Subgroup (2009) The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 25:2078–2079.
- [5] Li, H (2018) Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* 34:3094–3100.
- [6] Martin, KR, Le, HT, Abdelgawad, A, Yang, C, Lu, G, Keffer, JL, Zhang, X, Zhuang, Z, Asare-Okai, PN, Chan, CS, Batish, M, Yu, Y (2024) Development of an efficient, effective, and economical technology for proteome analysis. *Cell Reports Methods* 4:100796.
- [7] Sun, J, Xu, X, Wei, S, Yu, Y (2026) In-Cell Proteomics Enables High-Resolution Spatial and Temporal Mapping of Early *Xenopus tropicalis* Embryos. *Molecular & Cellular Proteomics* 25:101481.
- [8] Breuer, M, Earnest, TM, Merryman, C, Wise, KS, Sun, L, Lynott, MR, Hutchison, CA, Smith, HO, Lapek, JD, Gonzalez, DJ, De Crécy-Lagard, V, Haas, D, Hanson, AD, Labhsetwar, P, Glass, JI, Luthey-Schulten, Z (2019) Essential metabolism for a minimal cell. *eLife* 8:e36842.
- [9] Razin, S, Argaman, M, Avigan, J (1963) Chemical Composition of Mycoplasma Cells and Membranes. *Microbiology* 33:477–487.
- [10] Moger-Reischer, RZ, Glass, JI, Wise, KS, Sun, L, Bittencourt, DMC, Lehmkuhl, BK, Schoolmaster, DR, Lynch, M, Lennon, JT (2023) Evolution of a minimal cell. *Nature* 620:122–127.
- [11] Milo, R (2013) What is the total number of protein molecules per cell volume? A call to rethink some published values. *BioEssays* 35:1050–1055 _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/bies.201300066>.
- [12] Hallgren, J, Tsirigos, KD, Pedersen, MD, Almagro Armenteros, JJ, Marcatili, P, Nielsen, H, Krogh, A, Winther, O (2022) DeepTMHMM predicts alpha and beta transmembrane proteins using deep neural networks.
- [13] Teufel, F, Almagro Armenteros, JJ, Johansen, AR, Gíslason, MH, Pihl, SI, Tsirigos, KD, Winther, O, Brunak, S, Von Heijne, G, Nielsen, H (2022) SignalP 6.0 predicts all five types of signal peptides using protein language models. *Nat Biotechnol* 40:1023–1025.
- [14] Roots, CT, Lukasiewicz, A, Barrick, JE (2021) OSTIR: open source translation initiation rate prediction. *J Open Source Softw* 6:3362.

- [15] Sharp, PM, Li, WH (1987) The codon adaptation index-a measure of directional synonymous codon usage bias, and its potential applications. *Nucleic Acids Res* 15:1281–1295.
- [16] Burgos, R, Weber, M, Martinez, S, Lluch-Senar, M, Serrano, L (2020) Protein quality control and regulated proteolysis in the genome-reduced organism *Mycoplasma pneumoniae*. *Molecular Systems Biology* 16:e9530.
- [17] Maier, T, Schmidt, A, Güell, M, Kühner, S, Gavin, A, Aebersold, R, Serrano, L (2011) Quantification of mRNA and protein and integration with protein turnover in a bacterium. *Mol Syst Biol* 7:MSB201138.
- [18] Thermo Fisher Scientific (2016) TRIzol Reagent User Guide. Pub. No. MAN0001271, Rev. A.0.
- [19] Sandberg, TE, Wise, KS, Dalldorf, C, Szubin, R, Feist, AM, Glass, JI, Palsson, BO (2023) Adaptive evolution of a minimal organism with a synthetic genome. *iScience* 26:107500.